

Rule-Based Python Chatbot

30-Minute Practical Task

STUDENT NAME

Manoj Kumar.N

REGISTER NUMBER

99230041120

PROGRAMMING ENVIRONMENT

Python 3.10+

ASSESSMENT DOMAIN

Conversational Loops & Conditionals

1. ASSESSMENT OVERVIEW

The objective of this assignment is to develop an interactive rule-based conversational chatbot utilizing Python fundamentals. The implementation prompts for the user's name to maintain conversational context, runs within an infinite `while True:` loop, standardizes user utterances with `.lower().strip()`, evaluates intent using the `in` membership operator, answers specific academic and general queries, and cleanly breaks out when exit triggers are received.

- **Status:** All criteria verified — continuous loop execution, dynamic name response, multi-topic Q&A, default fallback handling, and terminal exit.

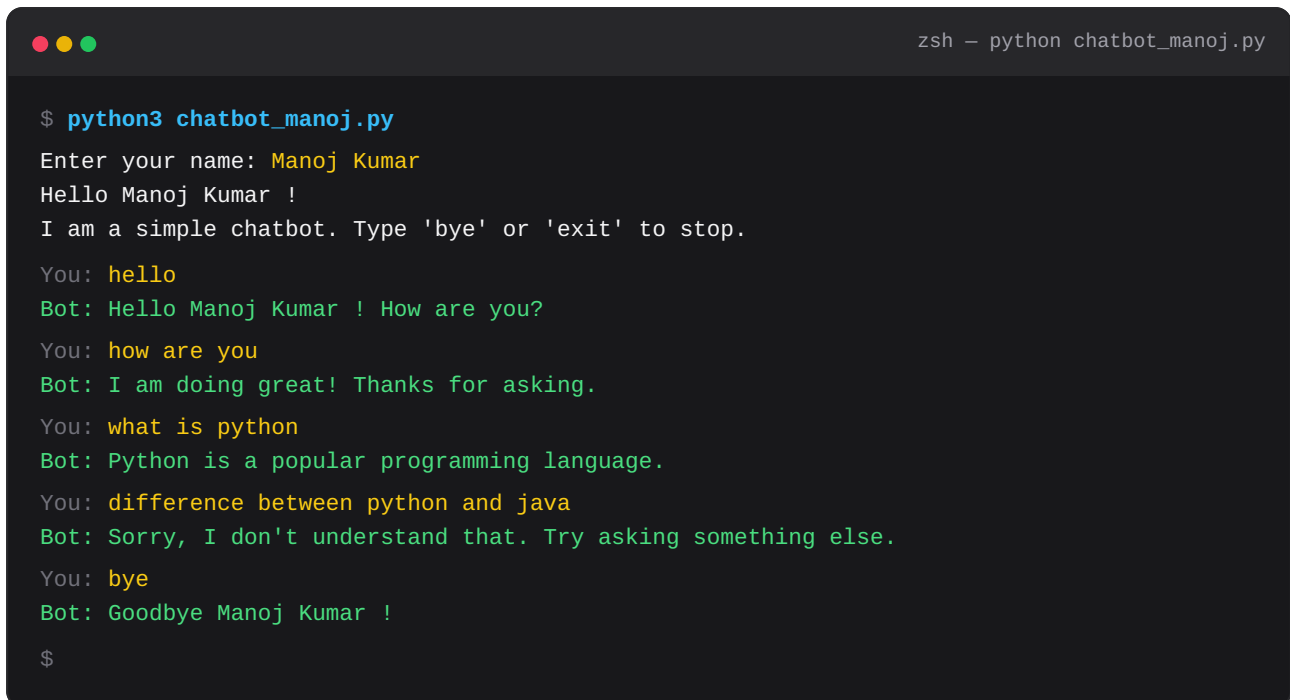
2. PYTHON IMPLEMENTATION

chatbot_manoj.py

UTF-8 • Python

```
1 # Rule-Based Conversational Chatbot
2 # Name: Manoj Kumar.N | Reg No: 99230041120
3 name = input("Enter your name: ")
4 print("Hello", name, "!")
5 print("I am a simple chatbot. Type 'bye' or 'exit' to stop.")
6
7 while True:
8     message = input("You: ").lower().strip()
9
10    if "hello" in message or "hi" in message:
11        print("Bot: Hello", name, "! How are you?")
12    elif "how are you" in message:
13        print("Bot: I am doing great! Thanks for asking.")
14    elif "what is python" in message:
15        print("Bot: Python is a popular programming language.")
16    elif "what can you do" in message:
17        print("Bot: I can answer simple questions and chat with you.")
18    elif "bye" in message or "exit" in message:
19        print("Bot: Goodbye", name, "!")
20        break
21    else:
22        print("Bot: Sorry, I don't understand that. Try asking something else.")
```

3. TERMINAL EXECUTION & OUTPUT VERIFICATION



```
zsh — python chatbot_manoj.py

$ python3 chatbot_manoj.py
Enter your name: Manoj Kumar
Hello Manoj Kumar !
I am a simple chatbot. Type 'bye' or 'exit' to stop.
You: hello
Bot: Hello Manoj Kumar ! How are you?
You: how are you
Bot: I am doing great! Thanks for asking.
You: what is python
Bot: Python is a popular programming language.
You: difference between python and java
Bot: Sorry, I don't understand that. Try asking something else.
You: bye
Bot: Goodbye Manoj Kumar !

$
```